Is Manipulation Hard for Successor Measures?

Aditya Mohan[1], **Caleb Chuck**[2], **Siddhant Agarwal**[2], **Amy Zhang**[2]
a.mohan@ai.uni-hannover.de

[1]Leibniz University Hannover
[2]University of Texas at Austin

Abstract

Unsupervised and goal-conditioned reinforcement learning methods often learn reusable future-reachability objects, such as goal-conditioned values, contrastive scores, successor measures, or predictive representations. We study a failure mode that arises when the observation or learned representation under-resolves local interaction variables. In the full physical state, the successor measure is well defined; the difficulty is that observation-near states can belong to different interaction regimes with different post-interaction futures. A learned score may therefore preserve coarse geometric reachability while weakly distinguishing the interaction variable that determines future controllability. We formalize this view through interaction-conditioned tail successor structure and instantiate it in the state-based version of OGBENCH `cube-single`, where the relevant interaction variable is whether the cube is freely moving, controllably held, slipping, or dropped. A phase diagnostic decomposes each rollout into approach, control acquisition, maintained control, and transport. Across five state-based future-reachability methods, approach failures are negligible, while most failures occur after control acquisition. Support diagnostics show that contact-regime coverage covaries with performance but cannot explain it: value-ranking probes on in-hand states drawn from the dataset itself, where coverage is adequate, show that the weaker methods' learned scores still separate those states only weakly. These results suggest that the missing object is not grasping-specific, but the preservation of interaction-conditioned successor structure.

1 Introduction

Unsupervised reinforcement learning (URL) promises reusable control: an agent learns a policy, representation, or value structure without task rewards and later adapts it to downstream goals. This is especially appealing in robotics, where the same workspace may contain many possible goals, but collecting new interaction data for every reward is expensive. URL methods can be understood as learning tractable approximations to policy-dependent successor measures, differing in how they parameterize, compress, or extract information about future state occupancy Agarwal et al. (2025a). Under this view, goal-conditioned values, contrastive state-goal scores, explicit successor-measure factorizations, and predictive representations provide different approximations to future occupancy (Eysenbach et al., 2021; 2022; Ma et al., 2022; Touati and Ollivier, 2021; Touati et al., 2023).

In many control problems, a local interaction state can sharply change which future states remain reachable. An object may be stably held or slipping; a door may be latched or unlatched; and a robot foot may be in stance or swing. At the level of the full physical state, each of these systems remains Markovian and has a well-defined successor measure. The difficulty arises when the observation or learned representation fails to distinguish interaction states with different futures. Two states may have nearly identical object and robot poses, for example, while one corresponds to stable control and the other to an incipient slip.

In this work, we investigate whether contact-rich manipulation exposes a failure mode in successor-measure approximations: learned representations or scores may preserve coarse geometric reachability while failing to distinguish interaction states that lead to different outcomes.

Hypothesis 1 (Interaction-state resolution)**.** *Successor-measure approximations can fail in manipulation when they assign similar representations or scores to nearby states with different futures. In cube manipulation, the relevant distinction is between states that support stable transport and states that lead to slipping or dropping.*

We formalize this discrepancy as a local successor gap: nearby states can have substantially different goal-relevant future occupancies, and therefore different probabilities of eventual task success.

We study this hypothesis in the state-based OGBENCH `cube-single` task Park et al. (2025). We decompose each rollout into four stages: reaching the cube, lifting it, retaining it at the end of the episode, and transporting it to the goal. This separates failures of coarse reachability from failures that occur after the object has been lifted. Across methods, failures rarely occur while reaching the cube, and most performance differences arise after the cube has been lifted. The lower-performing methods' scores also weakly rank in-hand states by whether the policy will retain the cube and complete the task. Dataset coverage covaries with performance but cannot account for the weak value ranking, which persists on in-hand states drawn from the dataset itself; coarse cube geometry likewise does not explain the ordering.

Our contributions are threefold: (i) We formulate interaction-state resolution as a potential failure mode of successor-measure approximations: nearby states can have different continuation futures while receiving similar learned representations or scores. (ii) We introduce a phase-based analysis of cube pick-and-place that separates reaching, lifting, final retention, and transport. (iii) We apply this analysis to five state-based methods on OGBENCH `cube-single` and find that performance differences arise primarily after the cube has been lifted. Support, geometric-decoding, and counterfactual value-ranking probes further show that although dataset coverage covaries with performance, it cannot explain the weak value ranking, which persists on in-hand states drawn from the dataset itself.

2 Preliminaries

We consider a reward-free Markov decision process $(\mathcal{S}, \mathcal{A}, P, \gamma)$ with state space $\mathcal{S}$, action space $\mathcal{A}$, transition kernel P, and discount factor γ. For a policy π, the successor measure from (s, a) to a measurable set $X \subseteq \mathcal{S}$ is:

$$M^\pi(s, a, X) = \mathbb{E}_\pi \left[\sum_{t \geq 0} \gamma^t \mathbf{1} s_{t+1} \in X \mid s_0 = s, a_0 = a \right]. \tag{1}$$

When the state space is discrete, we write $M^\pi(s, a, s^+)$ for the discounted occupancy of state s^+. For a state reward r, policy evaluation can be written as:

$$Q_r^\pi(s, a) = \int_{\mathcal{S}} M^\pi(s, a, ds^+) r(s^+). \tag{2}$$

This identity motivates successor measures for URL: a reward-independent representation of future occupancy can be combined with a downstream reward to evaluate task-specific policies (Agarwal et al., 2025a; Dayan, 1993; Touati and Ollivier, 2021; Touati et al., 2023).

Representing exact successor measures is intractable in large or continuous state spaces, particularly when the representation must cover many policies. Practical URL methods therefore impose additional structure. For example, FB uses a low-rank approximation:

$$M^{\pi_z}(s, a, ds^+) \approx F(s, a, z)^\top B(s^+) \, \rho(ds^+), \tag{3}$$

where ρ is a reference state distribution and z indexes a reward or policy embedding (Touati and Ollivier, 2021; Touati et al., 2023). Other methods can be viewed as learning lower-dimensional or goal-specific projections of future occupancy (Agarwal et al., 2025a). Goal-conditioned methods learn values or policies indexed by a target, contrastive methods learn state–goal scores, and predictive methods learn latent transition structure intended to support long-horizon control (Eysenbach et al., 2021; 2022; Ma et al., 2022). Each objective therefore determines which state distinctions are preserved for downstream control.

2.1 Successor-measure view of the evaluated methods

Let $o_G(s)$ denote the task-relevant goal coordinates extracted from state s. For a goal g, define the success set

$$G(g) = \left\{ s^+ : \left\| o_G(s^+) - g \right\| \leq \epsilon_G \right\}. \tag{4}$$

The corresponding goal-relevant successor mass is

$$m_g^\pi(s, a) = M^\pi(s, a, G(g)). \tag{5}$$

This quantity measures the discounted future occupancy of the goal set under π. We use m_g^π only as a common reference: the evaluated methods do not all estimate it explicitly.

Under this view, GCIQL learns a goal-conditioned value $Q(s, a, g)$ (Park et al., 2025), while CRL learns a bilinear state-action-goal score $f(s, a)^\top b(g)$ (Eysenbach et al., 2022). FB directly learns a low-rank factorization of policy-conditioned successor measures (Touati and Ollivier, 2021; Touati et al., 2023), and PSM learns a basis for representing a family of successor measures (Agarwal et al., 2025b). RLDP instead learns predictive features from which downstream task-conditioned policies are extracted (Jajoo et al., 2026).

Contact-rich manipulation can violate the assumption that geometrically similar states have similar futures. Nearby robot-object configurations may correspond to a stable grasp, an incipient slip, or a loss of control, and can therefore induce substantially different future occupancies. A useful learned representation or score must consequently preserve distinctions between interaction-states that affect future control, in addition to the coarse geometry of the robot, object, and goal.

3 Interaction-state resolution in successor measures

We now formalize the interaction-state hypothesis behind our analysis. The exact successor measure remains well defined in the full physical state, but a learned representation or score may fail to distinguish interaction states with different future occupancies.

3.1 Full-state successor measures under learned representations

A full-state successor measure can represent the interaction mode since the interaction state is part of the full physical state. The error arises when function approximation is introduced to learn state features. Let $y = \Omega(\tilde{s})$ denote the observation used by the policy or value function, and let $\omega(\tilde{s})$ denote the local interaction state, such as whether an object is freely moving, stably held, or slipping. Two full states can be nearby in observation space while having different interaction states and therefore different future occupancies. The relevant question is therefore which interaction-state distinctions are lost in the observation, learned representation, or scalar score. We illustrate this distinction by decomposing the transition dynamics according to interaction state.

3.2 Interaction-dependent transition dynamics

To illustrate how interaction state changes future occupancy, consider a binary projection $\omega(\tilde{s}) \in \{0, 1\}$. Here, $\omega = 0$ denotes a passive or pre-contact state, while $\omega = 1$ denotes a state in which

the robot action directly affects the object. Let r_t denote the robot state and o_t the object state. For illustration, write the object transition as

$$o_{t+1} \sim \begin{cases} P_0(\cdot \mid o_t), & \omega_t = 0, \\ P_1(\cdot \mid r_t, o_t, a_t), & \omega_t = 1. \end{cases} \tag{6}$$

This binary projection records whether the robot action directly affects the object; finer interaction states may distinguish stable control, slipping, and loss of control. When $\omega_t = 0$, the object transition does not depend directly on the robot action. When $\omega_t = 1$, it depends on the robot state and action. Thus, similar robot–object configurations can have different successor measures when their interaction states induce different object dynamics.

3.3 Control acquisition and task success

In interaction-rich tasks, acquiring control of the object separates reaching it from completing the task. A policy may satisfy a coarse control-acquisition criterion while reaching a state from which retaining and transporting the object remains unlikely.

Let $C \subseteq \widetilde{\mathcal{S}}$ denote the control-acquisition set and define its first hitting time as $\tau_C = \inf\{t : \tilde{s}_t \in C\}$. In cube manipulation, states in C may correspond to either a stable grasp or an unstable grasp that is likely to slip. The event $\tau_C \leq H$ is therefore not sufficient for task success: success also depends on the state from which the policy begins transporting the cube.

For a remaining horizon k, define the task-success value from state $\tilde{s}$:

$$v_k^\pi(\tilde{s}; g) = \Pr_\pi\left(\tilde{s}_k \in R \cap G_g \mid \tilde{s}_0 = \tilde{s}\right). \tag{7}$$

Conditioning on the first control-acquisition time gives

$$u_H^\pi(\tilde{s}, a; g) = \mathbb{E}_\pi\left[\mathbf{1}\{\tau_C \leq H\} v_{H-\tau_C}^\pi(\tilde{s}_{\tau_C}; g) \mid \tilde{s}_0 = \tilde{s}, a_0 = a\right]. \tag{8}$$

This decomposition separates whether the policy acquires control from whether it completes the task after control has been acquired.

Our experiments show that most performance differences arise after control acquisition. Policies generally reach and lift the cube, but many fail to retain it or transport it to the goal. Stable and unstable grasp states can both satisfy the same lifting criterion while assigning future occupancy to different outcomes. A stable grasp places more future occupancy on retained-cube and goal states, whereas an unstable grasp places more occupancy on dropped-cube or repeated-pickup states.

3.4 Local successor gaps in learned scores

For an observation scale $\epsilon > 0$ and remaining horizon k, define the local successor gap

$$\Gamma_{\epsilon,k}^\pi(g) = \sup_{\substack{\tilde{s}_1, \tilde{s}_2 \in C \\ \|\Omega(\tilde{s}_1) - \Omega(\tilde{s}_2)\| \leq \epsilon}} \left| v_k^\pi(\tilde{s}_1; g) - v_k^\pi(\tilde{s}_2; g) \right|. \tag{9}$$

A large $\Gamma_{\epsilon,k}^\pi(g)$ means that nearby states reached after lifting have substantially different task-success values. Such a gap arises when two states satisfy the same lifting criterion, but only one is likely to retain the cube and reach the goal.

Suppose a stable grasp state $\tilde{s}_s$ and an unstable grasp state $\tilde{s}_u$ produce the same observation,

$$\Omega(\tilde{s}_s) = \Omega(\tilde{s}_u) = y,$$

but have different task-success values:

$$v_s = v_k^\pi(\tilde{s}_s; g), \qquad v_u = v_k^\pi(\tilde{s}_u; g).$$

If p is the probability that observation y corresponds to the stable grasp state, then the optimal observation-only predictor under squared error is

$$\bar{v}(y) = pv_s + (1-p)v_u. \tag{10}$$

Its errors on the two modes are

$$|v_s - \bar{v}| = (1-p)|v_s - v_u|, \qquad |v_u - \bar{v}| = p|v_s - v_u|. \tag{11}$$

Thus, if stable and unstable grasp states receive the same observation or representation, an observation-based value estimate must underestimate the stable state, overestimate the unstable state, or both. Overestimating unstable grasp states can cause the actor to prefer actions that lift the cube but do not maintain control during transport.

This information loss may occur either in the learned representation or in the scalar readout. A bilinear score can express the distinction only if the representation encodes the relevant interaction-state information and the task-conditioned readout uses it. Our empirical probes therefore ask whether the learned representation contains information about task success from lifted-cube states and whether the scalar score preserves that information.

3.5 Instantiation in cube manipulation

We study this hypothesis in the state-based OGBENCH `cube-single` environment. Here, the relevant interaction state describes whether the cube is freely moving, stably held, slipping, or no longer controlled. The observation contains the gripper pose, cube pose, gripper opening, and contact indicators, but the distinction between stable and unstable grasps is distributed across these state variables. Consequently, nearby robot–cube configurations can differ in whether the policy will retain the cube during transport.

For goal g, we use the task-specific event

$$\varphi_g(\tilde{s}_{0:H}) = \mathbf{1}\{\tau_A \le H,\ \tau_C \le H,\ \tilde{s}_H \in R,\ \tilde{s}_H \in G_g\}, \tag{12}$$

where A is the reaching set, C is the control-acquisition set, R is the final-retention set, and G_g is the goal set. The corresponding task-success probability is

$$u_H^\pi(\tilde{s}, a; g) = \Pr_\pi\left(\tau_A \le H,\ \tau_C \le H,\ \tilde{s}_H \in M,\ \tilde{s}_H \in G_g \mid \tilde{s}_0 = \tilde{s}, a_0 = a\right). \tag{13}$$

The four criteria distinguish reaching the cube, lifting it, retaining it at the end of the episode, and transporting it to the goal. A final-retention failure means that the policy satisfied the lifting criterion but did not end the episode with the cube above the retention threshold.

4 Diagnostic Method

We now turn the interaction-state hypothesis from Section 3 into a rollout-based analysis. The goal is to separate failures in reaching and lifting the cube from failures that occur afterward, when the policy must retain and transport it.

4.1 Evaluation setting

We study the state-based version of OGBENCH `cube-single` over five held-out evaluation goals, denoted *task1* - *task5*. Using state observations removes visual representation learning as a confound and focuses the analysis on state-based value and policy learning. We evaluate five state-based URL methods: GCIQL, CRL, FB, RLDP, and PSM. We report interquartile mean (IQM) (Agarwal et al., 2021) success with stratified-bootstrap confidence intervals for aggregate performance. We then analyze deterministic closed-loop rollouts using the stage-based decomposition below.

4.2 Rollout-stage decomposition

We instantiate the sets from Section 3 using the simulator-state thresholds in Table 1. Each rollout is successful if all criteria are satisfied; otherwise, it is assigned to the first unmet criterion in the ordered sequence. This converts aggregate success into a stage-specific failure profile.

Table 1: Operational criteria for assigning the first unmet stage in state-based rollouts.

Stage	Criterion
Reached cube	Minimum gripper-to-cube distance < 6 cm at any point in the episode
Lifted cube	Cube lifted > 3 cm above the table with the gripper closed for at least 5 consecutive steps
Final retention	Final cube height above the table ≥ 3 cm
Goal reached	Final cube pose within 5 cm of the goal pose

The criteria are computed directly from simulator states and applied identically to every policy. They do not depend on a method's learned value, representation, score, or model estimate. The same decomposition can therefore be applied to goal-conditioned, contrastive, predictive, successor-measure, and model-based methods.

4.3 Cross-method failure profile

For each method, we report the percentage of episodes assigned to success, reaching failure, lifting failure, final-retention failure, and goal-reaching failure. This distribution distinguishes methods that fail to reach or lift the cube from methods that lift it but fail to retain it or reach the goal. We also report the post-lift share of failures:

$$\text{Post-lift share} = \frac{\text{Final-retention fail} + \text{Goal-reaching fail}}{100 - \text{Success}}. \tag{14}$$

This quantity measures the fraction of failed episodes in which the cube was lifted before the rollout failed.

5 Results

Our experiments test the interaction-state hypothesis from Section 3. The evaluated methods rarely fail during approach; most performance differences arise after control has been acquired. We then ask whether this pattern is explained by state-space support and whether learned scores distinguish acquired-control states with different task outcomes. These diagnostics identify where policies fail and whether learned values preserve the interaction-state distinctions needed for control, but they are not designed to isolate a single architectural cause. Per-task success, exact counts, and the full diagnostic tables are in Appendix A.

5.1 Aggregate success obscures where policies fail

Aggregate state-based success ranks the methods but says nothing about where they fail. Across 10 evaluation seeds, the overall interquartile mean (IQM) (Agarwal et al., 2021) success rates are GCIQL (71.8 ± 4.1), FB (52.0 ± 5.5), PSM (36.7 ± 4.5), CRL (25.8 ± 2.3), and RLDP (22.7 ± 6.0).[1]

[1] Aggregate success uses the stochastic evaluation policy and is summarized by IQM over seeds; the phase decomposition uses deterministic closed-loop rollouts, so per-method success differs slightly between the two analyses. We report each metric consistently within its corresponding analysis.

The rollout-stage decomposition in Section 4 localizes where the methods fail (Figures 1 and 2). Every method almost always completes the approach phase, and most rollouts satisfy the control-acquisition criterion. The largest differences arise afterward, in maintain-control and transport failures.

FB and GCIQL differ little during approach and control acquisition. Relative to GCIQL, FB has 17.1 percentage points more maintain-control failures and 10.3 points more transport failures, partly offset by 3.2 points fewer control-acquisition failures. The same pattern appears for other methods: PSM and RLDP acquire control but fail to maintain it in 36% and 44% of episodes, respectively. Across the lower-performing methods, 65–83% of failed episodes occur after the control-acquisition criterion has been satisfied (Appendix A). Because the phase labels are computed only from simulator states, the same analysis can be applied to other methods without access to their learned values, representations, or models.

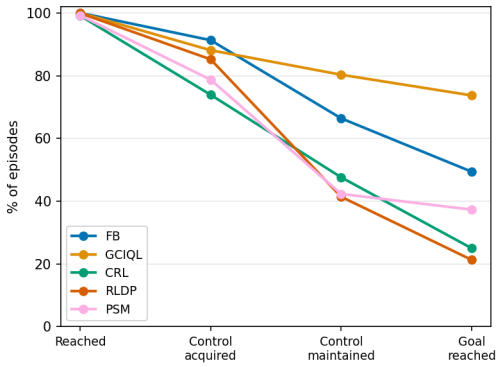


Figure 1: **Phase survival through cube manipulation.** Each curve shows the fraction of deterministic evaluation rollouts that satisfy each successive phase criterion: approach, control acquisition, maintain control, and transport.

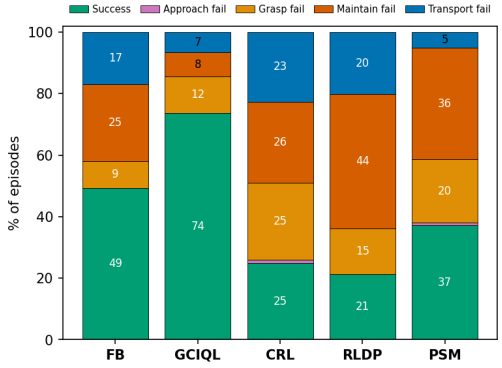


Figure 2: **Terminal outcomes under the phase decomposition.** Each rollout is assigned to success or to the first unmet phase criterion. Approach failures are negligible across methods, while most failures occur in maintain-control and transport.

5.2 Dataset support covaries with performance but cannot explain it

The support analysis groups individual states according to gripper and cube state. Because both the play data and policy rollouts contain repeated pick-and-place attempts, we use three state categories: *approach* (pre-contact), *pickup* (gripper closed, cube on the table), and *carry* (cube lifted and held). These categories are distinct from the terminal failure phases: maintain-control and transport failures are assigned according to how the episode ends, whereas approach, pickup, and carry describe states encountered during the rollout.

One possible explanation is that failures after control acquisition occur in regions that are poorly represented in the offline data. Figure 3 reports the difference in nearest-neighbor distance to the offline dataset between failed and successful rollout states, grouped by state category and failure phase. Failed states are often farther from the data in the pickup and carry categories, especially for maintain-control failures (e.g. CRL pickup, $+1.83$).

These support differences track the ordering of method success. GCIQL, the strongest method, shows essentially no gap: its failed states are as close to the data as its successful ones (all categories within 0.1). Every lower-performing method shows a positive gap. Dataset support is therefore correlated with performance under this state-only metric, and we cannot rule it out as a contributing factor. It does not, however, explain the value-ranking failures reported below. That probe resets to in-hand states drawn *from the offline data*, where support is adequate by construction, yet the

lower-performing methods still rank those states near chance. A consistent reading is that the two are linked: a value that does not distinguish stable from unstable grasps produces actions that drive the policy into states the dataset does not cover, so the support gap is a consequence of weak value ranking rather than a competing explanation for it. This analysis does not rule out differences in action-conditioned support, transition support, or the availability of corrective actions in the offline data.

5.3 Value functions weakly rank acquired-control states

Because dataset support cannot account for ranking failures on states drawn from the dataset itself, we next investigate whether each learned score ranks in-hand grasp states by eventual success under the method's own policy. Coarse cube-to-goal geometry is not sufficient to explain performance. For methods with an explicit backward embedding, the embeddings learned by PSM and RLDP predict cube-to-goal distance substantially better than the embedding learned by FB (nonlinear R^2 of 0.90 and 0.75 versus 0.21). Nevertheless, both methods have higher maintain-control failure rates than FB (Appendix A). Encoding cube-to-goal geometry therefore does not guarantee that the policy will maintain control.

We evaluate value quality through simulator resets (Figure 4). We reset the simulator to 200 in-hand grasp states drawn from the offline data. From each state, we roll out the agent's own policy and label the grasp by whether the rollout reaches the goal. Each grasp is evaluated independently, and the probe covers states from the offline dataset rather than only states generated by the learned policy. We then plot success rate as a function of the learned value percentile. We summarize ranking quality using rank-AUC, the probability that a successful grasp state receives a higher value than an unsuccessful grasp state. GCIQL's value is strongly predictive (rank-AUC 0.89): its top-value grasps succeed $\approx 99\%$ of the time, its bottom-value grasps $\approx 27\%$. For RLDP and PSM, the curves are nearly flat (AUC 0.52 and 0.49), indicating that higher values do not correspond to higher success rates. FB shows a weaker relationship (AUC approximately 0.61), while CRL is intermediate (0.68). For FB and RLDP, the learned value is therefore only weakly informative about whether the policy will maintain control and complete the task.

We next ask whether the weak ranking arises in the learned representation or in the scalar value derived from it. Using each agent's simulator-reset success labels, we fit a linear readout from the learned representation and compare its ranking with that of the learned value (Appendix A, Figure 5). The probe suggests different bottlenecks for FB and RLDP. FB is consistent with a readout bottleneck: a linear readout of its forward map reaches AUC 0.70, while its learned scalar value reaches 0.61. RLDP is consistent with a representation bottleneck: the best readout reaches only 0.61, and its learned value reaches 0.55. For GCIQL, the learned value preserves nearly all of the linearly decodable signal available in its value features, reaching AUC 0.89.

These probes suggest that weak ranking can arise either in the learned representation or in the scalar readout, but they do not identify the readout form as the sole cause. GCIQL differs from FB, CRL, and RLDP in objective, actor training, critic architecture, implicit regularization, and optimization stability, not only in its value readout. The evidence therefore supports a narrower conclusion: the lower-performing methods learn scores that weakly rank acquired-control states by whether the agent will maintain control and complete transport, while GCIQL provides a stronger ranking under this diagnostic. A matched readout ablation would be needed to attribute the difference specifically to bilinear versus MLP value form.

Takeaway. The phase decomposition changes the interpretation of state-based URL failures in cube manipulation. The evaluated methods usually complete approach and acquire control, but many failures occur afterward, in the maintain-control and transport phases. Failed pickup and carry states are farther from the offline dataset for the lower-performing methods but not for GCIQL, so support covaries with performance; it cannot, however, explain why the learned values rank grasp states drawn from the dataset itself near chance. The learned scores of the lower-performing methods also

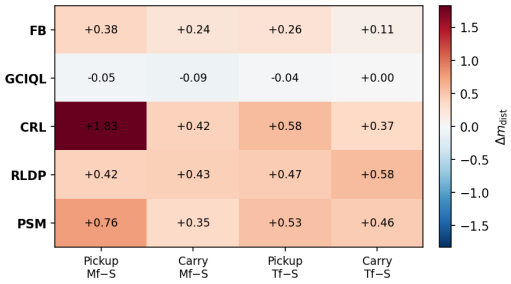


Figure 3: **Failure-success support gap.** Positive values indicate that failed rollout states are farther from the offline dataset under the nearest-neighbor metric. The gap is positive for every lower-performing method but is essentially zero for GCIQL, the strongest method, so support covaries with performance.

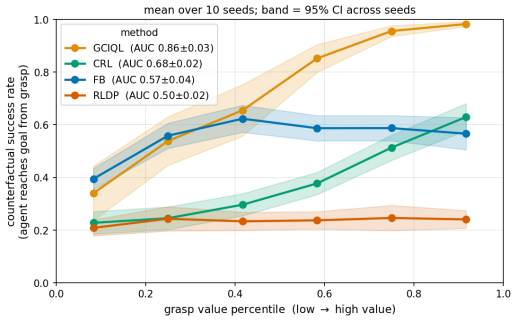


Figure 4: **Counterfactual value test.** For 200 in-hand grasps reset into the simulator and rolled out under each agent's policy, success rate as a function of the grasp state's value percentile.

do not reliably rank acquired-control states by whether the policy will maintain control and complete transport. Distinguishing such states is therefore a useful target for future offline URL objectives in manipulation.

6 Related Work

Unsupervised and goal-conditioned RL. URL methods learn reusable control objects in a reward-free or reward-agnostic phase and reuse them for downstream goals. Goal-conditioned methods learn values or policies indexed by target states (Eysenbach et al., 2021; Ma et al., 2022), while contrastive and metric-based methods learn state-goal scores or representations that distinguish future states from negatives (Eysenbach et al., 2022; Park et al., 2024; Wang et al., 2023). Successor representations, successor features, and FB representations estimate future-occupancy objects that can be combined with downstream rewards (Barreto et al., 2017; Dayan, 1993; Touati and Ollivier, 2021; Touati et al., 2023). Proto-value and proto-successor methods instead use learned bases or spectral structure to represent behavior or value spaces (Agarwal et al., 2025b; Farebrother et al., 2023; Mahadevan, 2005; Mahadevan and Maggioni, 2007). Recent work has argued that many URL objectives can be viewed as tractable approximations to successor measures or future-reachability objects (Agarwal et al., 2025a). Our work uses this view diagnostically: we ask whether these learned objects preserve local interaction modes that change successor structure, rather than proposing a new URL objective.

Structured and controllable representations. Classical factored MDP work exploits state structure to make planning and learning more efficient (Boutilier et al., 1995; Guestrin et al., 2003; Kearns and Koller, 1999); see also Mohan et al. (2024). More recent controllable-representation methods aim to preserve action-relevant state information, often through inverse or multi-step inverse dynamics objectives (Efroni et al., 2022; Lamb et al., 2022; Levine et al., 2024; Rudolph et al., 2024). These works are closely related to our interaction-mode view: the relevant abstraction should retain variables that change how actions affect future states. Our contribution is different in scope. We do not learn a new abstraction or assume a known factorization; instead, we diagnose whether existing state-based URL scores collapse observation-near states whose long-horizon futures differ because of local interaction variables.

Contact-rich dynamics and manipulation. Contact-rich manipulation is a canonical setting where local interaction variables change effective dynamics. Classical work on pushing and manipulation

emphasizes that contact mode, friction, and stability determine controllability and feasible motion (Lynch and Mason, 1996; Mason, 1986). Trajectory-optimization and contact-implicit methods explicitly model or optimize through contact transitions (Aceituno-Cabezas and Rodriguez, 2020; Jin, 2024; Manchester and Kuindersma, 2017; Patel et al., 2019; Posa et al., 2014). Learning-based contact models also study the difficulty of representing discontinuous or hybrid contact dynamics (Hogan and Rodriguez, 2016; Pfrommer et al., 2020). Our paper does not propose a contact model. Instead, we use contact-rich cube manipulation as a controlled diagnostic for a broader URL question: whether learned future-reachability scores preserve interaction-conditioned successor distinctions.

Offline goal-conditioned benchmarks and diagnostics. Offline goal-conditioned RL evaluates whether policies or representations learned from static data can solve downstream goals without additional online collection (Kostrikov et al., 2022; Ma et al., 2022; Park et al., 2025). Benchmarks such as OGBENCH report aggregate goal-reaching success, which is useful for comparing algorithms but does not identify the physical stage at which a policy fails (Park et al., 2025). Our diagnostic decomposes manipulation rollouts into approach, control acquisition, maintained control, and transport. This separates coarse reachability from the local interaction transition and from the ability to preserve and exploit that interaction regime.

7 Conclusion

We studied where state-based future-reachability methods fail in contact-rich cube manipulation. Our first step was to decompose each rollout into approach, control acquisition, maintained control, and transport. Across five methods, aggregate success hides a consistent failure profile: policies rarely fail to reach the cube and often acquire coarse control, but many failures occur *after* acquisition, when the post-interaction tail must preserve control and reach the goal. Support diagnostics show that state-space coverage covaries with performance but does not explain it: on in-hand states drawn from the dataset, where coverage is adequate, the weaker methods' values still rank near chance. Value-ranking diagnostics further suggest that learned scores only weakly distinguish in-hand post-acquisition states with different post-interaction futures. These results point to interaction-conditioned successor structure as a useful target for future offline goal-conditioned and reward-free control methods. The goal is not only to encode where the robot, object, and goal are, but also to preserve the interaction-mode distinctions that determine which tail successor regime follows a local interaction event.

References

B. Aceituno-Cabezas and A. Rodriguez. A global quasi-dynamic model for contact-trajectory optimization in manipulation. In *Robotics: Science and Systems*, 2020.

R. Agarwal, M. Schwarzer, P. Samuel Castro, A. C. Courville, and M. G. Bellemare. Deep reinforcement learning at the edge of the statistical precipice. In Ranzato et al. (2021).

S. Agarwal, C. Chuck, H. Sikchi, J. Hu, M. Rudolph, S. Niekum, P. Stone, and A. Zhang. A unified framework for unsupervised reinforcement learning algorithms. In *RLBrew: Ingredients for Developing Generalist Agents Workshop (RLC 2025)*, 2025a.

S. Agarwal, H. Sikchi, P. Stone, and A. Zhang. Proto successor measure: Representing the behavior space of an rl agent. In *Proceedings of the 42nd International Conference on Machine Learning (ICML'25)*, 2025b.

A. Barreto, W. Dabney, R. Munos, J. J. Hunt, T. Schaul, H. P. van Hasselt, and D. Silver. Successor features for transfer in reinforcement learning. In I. Guyon, U. von Luxburg, S. Bengio, H. Wallach, R. Fergus, S. Vishwanathan, and R. Garnett, editors, *Proceedings of the 31st International Conference on Advances in Neural Information Processing Systems (NeurIPS'17)*. Curran Associates, 2017.

C. Boutilier, R. Dearden, and M. Goldszmidt. Exploiting structure in policy construction. In C. Mellish, editor, *Proceedings of the 14th International Joint Conference on Artificial Intelligence (IJCAI'95)*. Morgan Kaufmann Publishers, 1995.

P. Dayan. Improving generalization for temporal difference learning: The successor representation. *Neural Computation*, 1993.

Y. Efroni, D. Misra, A. Krishnamurthy, A. Agarwal, and J. Langford. Provably filtering exogenous distractors using multistep inverse dynamics. In *The Tenth International Conference on Learning Representations (ICLR'22)* ICL (2022). Published online: iclr.cc.

B. Eysenbach, R. Salakhutdinov, and S. Levine. C-learning: Learning to achieve goals via recursive classification. In *The Ninth International Conference on Learning Representations (ICLR'21)*. ICLR, 2021. Published online: iclr.cc.

B. Eysenbach, T. Zhang, S. Levine, and R. Salakhutdinov. Contrastive learning as goal-conditioned reinforcement learning. In Koyejo et al. (2022).

J. Farebrother, J. Greaves, R. Agarwal, C. L. Lan, R. Goroshin, P. S. Castro, and M. G. Bellemare. Proto-value networks: Scaling representation learning with auxiliary tasks. In *The Eleventh International Conference on Learning Representations (ICLR'23)* ICL (2023). Published online: iclr.cc.

C. Guestrin, D. Koller, R. Parr, and S. Venkataraman. Efficient solution algorithms for factored mdps. *Journal of Artificial Intelligence Research*, 2003.

F. Hogan and A. Rodriguez. Feedback control of the pusher-slider system: A story of hybrid and underactuated contact dynamics. *arXiv preprint arXiv:1611.08268*, 2016.

Proceedings of the International Conference on Learning Representations (ICLR'22), 2022. ICLR. Published online: iclr.cc.

Proceedings of the International Conference on Learning Representations (ICLR'23), 2023. ICLR. Published online: iclr.cc.

P. Jajoo, H. Sikchi, S. Agarwal, A. Zhang, S. Niekum, and M. White. Regularized latent dynamics prediction is a strong baseline for behavioral foundation models. In *The Fourteenth International Conference on Learning Representations*, 2026.

W. Jin. Complementarity-free multi-contact modeling and optimization for dexterous manipulation. *arXiv preprint arXiv:2408.07855*, 2024.

M. Kearns and D. Koller. Efficient reinforcement learning in factored mdps. In T. Dean, editor, *Proceedings of the 16th International Joint Conference on Artificial Intelligence (IJCAI'99)*. Morgan Kaufmann Publishers, 1999.

I. Kostrikov, A. Nair, and S. Levine. Offline reinforcement learning with implicit q-learning. In *The Tenth International Conference on Learning Representations (ICLR'22)* ICL (2022). Published online: iclr.cc.

S. Koyejo, S. Mohamed, A. Agarwal, D. Belgrave, K. Cho, and A. Oh, editors. *Proceedings of the 36th International Conference on Advances in Neural Information Processing Systems (NeurIPS'22)*, 2022. Curran Associates.

A. Lamb, R. Islam, Y. Efroni, A. Didolkar, D. Misra, D. Foster, L. Molu, R. Chari, A. Krishnamurthy, and J. Langford. Guaranteed discovery of control-endogenous latent states with multi-step inverse models. *arXiv preprint arXiv:2207.08229*, 2022.

A. Levine, P. Stone, and A. Zhang. Multistep inverse is not all you need. *arXiv preprint arXiv:2403.11940*, 2024.

K. Lynch and M. Mason. Stable pushing: Mechanics, controllability, and planning. *The International Journal of Robotics Research*, 1996.

Y. J. Ma, J. Yan, Dinesh Jayaraman, and O. Bastani. Offline goal-conditioned reinforcement learning via f-advantage regression. In Koyejo et al. (2022).

S. Mahadevan. Proto-value functions: Developmental reinforcement learning. In L. Raedt and S. Wrobel, editors, *Proceedings of the 22nd International Conference on Machine Learning (ICML'05)*. Omnipress, 2005.

S. Mahadevan and M. Maggioni. Proto-value functions: A laplacian framework for learning representation and control in markov decision processes. *Journal of Machine Learning Research*, 8: 2169–2231, 2007.

Z. Manchester and S. Kuindersma. Variational contact-implicit trajectory optimization. In *International Symposium on Robotics Research*, 2017.

M. Mason. Mechanics and planning of manipulator pushing operations. *The International Journal of Robotics Research*, 1986.

A. Mohan, A. Zhang, and M. Lindauer. Structure in deep reinforcement learning: A survey and open problems. *Journal of Artificial Intelligence Research*, 79, 2024.

S. Park, T. Kreiman, and S. Levine. Foundation policies with hilbert representations. In R. Salakhutdinov, Z. Kolter, K. Heller, A. Weller, N. Oliver, J. Scarlett, and F. Berkenkamp, editors, *Proceedings of the 41st International Conference on Machine Learning (ICML'24)*, volume 251 of *Proceedings of Machine Learning Research*. PMLR, 2024.

S. Park, K. Frans, B. Eysenbach, and S. Levine. Ogbench: Benchmarking offline goal-conditioned rl. In *The Thirteenth International Conference on Learning Representations (ICLR'25)*. ICLR, 2025. Published online: iclr.cc.

A. Patel, S. Shield, S. Kazi, A. Johnson, and L. Biegler. Contact-implicit trajectory optimization using orthogonal collocation. *IEEE Robotics and Automation Letters*, 2019.

S. Pfrommer, M. Halm, and M. Posa. ContactNets: Learning discontinuous contact dynamics with smooth, implicit representations. In J. Kober, F. Ramos, and C. Tomlin, editors, *Proceedings of the 4th Annual Conference on Robot Learning (CoRL'20)*, volume 155. PMLR, 2020.

M. Posa, C. Cantu, and R. Tedrake. A direct method for trajectory optimization of rigid bodies through contact. *The International Journal of Robotics Research*, 2014.

M. Ranzato, A. Beygelzimer, K. Nguyen, P. Liang, J. Vaughan, and Y. Dauphin, editors. *Proceedings of the 35th International Conference on Advances in Neural Information Processing Systems (NeurIPS'21)*, 2021. Curran Associates.

M. Rudolph, C. Chuck, K. Black, M. Lvovsky, S. Niekum, and A. Zhang. Learning action-based representations using invariance. In *Reinforcement Learning Conference*, 2024.

A. Touati and Y. Ollivier. Learning one representation to optimize all rewards. In Ranzato et al. (2021).

A. Touati, J. Rapin, and Y. Ollivier. Does zero-shot reinforcement learning exist? In *The Eleventh International Conference on Learning Representations (ICLR'23)* ICL (2023). Published online: iclr.cc.

T. Wang, A. Torralba, P. Isola, and A. Zhang. Optimal goal-reaching reinforcement learning via quasimetric learning. In A. Krause, E. Brunskill, K. Cho, B. Engelhardt, S. Sabato, and J. Scarlett, editors, *Proceedings of the 40th International Conference on Machine Learning (ICML'23)*, volume 202 of *Proceedings of Machine Learning Research*. PMLR, 2023.

A Additional results

This appendix provides the detailed numbers behind the main-paper diagnostics: per-task success, the exact terminal-outcome distribution, the full dataset-support distances, the backward-embedding geometry probe, and the counterfactual value/representation-ranking analysis. These results support the main empirical chain: scalar success hides post-acquisition failures; state-space support covaries with performance but cannot explain ranking failures on states drawn from the dataset itself; coarse cube-goal geometry is insufficient; and learned values differ in how well they rank in-hand states by the agent's own future success.

A.1 Per-task success

Table 2: State-based evaluation success by task (IQM $\pm$ half-width of the 95% stratified-bootstrap CI across 10 seeds). All methods use the flow-Q learning actor with BC regularization and are evaluated at 1M steps.

Method	task1	task2	task3	task4	task5	Overall
FB	21.7 ± 10.8	73.3 ± 10.8	73.3 ± 6.7	36.7 ± 15.8	50.0 ± 13.3	52.0 ± 5.5
GCIQL	78.8 ± 8.0	74.8 ± 5.8	76.4 ± 7.4	67.2 ± 4.2	68.0 ± 5.2	71.8 ± 4.1
CRL	23.6 ± 3.2	26.8 ± 7.4	28.4 ± 5.6	23.6 ± 3.8	27.2 ± 5.4	25.8 ± 2.3
RLDP	28.3 ± 10.8	16.7 ± 6.7	35.0 ± 15.9	13.3 ± 12.5	18.3 ± 10.0	22.7 ± 6.0
PSM	50.0 ± 11.7	38.3 ± 16.7	73.3 ± 12.5	1.7 ± 4.2	11.7 ± 9.2	36.7 ± 4.5

A.2 Terminal-outcome distribution

Table 3: Cross-method terminal-outcome distribution (% of episodes, mean across 10 seeds; the first five columns sum to 100%). The post-contact share is $(\text{maintain} + \text{transport})/(100 - \text{success})$.

Method	Success	Approach	Grasp	Maintain	Transport	Post-contact share
FB	49.4	0.0	8.7	24.9	17.0	82.8%
GCIQL	73.7	0.0	11.9	7.8	6.7	54.9%
CRL	25.0	0.9	25.2	26.3	22.6	65.2%
RLDP	21.3	0.1	14.7	43.8	20.1	81.2%
PSM	37.3	0.8	20.5	36.4	5.0	66.0%

A.3 Dataset-manifold support

Table 4: Nearest-neighbor distance m_{dist} from rollout states to comparable offline states (mean across seeds; larger means lower support), by per-state regime and terminal outcome. S = success, Mf = maintain-control failure, Tf = transport failure. All methods use the flow actor and are evaluated at 1M steps under the same probe (20 episodes per seed, 10 seeds). Figure 3 plots the Mf−S and Tf−S deltas for the contact regimes.

Method	Regime	S	Mf	Tf
GCIQL	approach	2.53	2.29	2.43
	pickup	3.05	3.00	3.01
	carry	2.67	2.58	2.67
FB	approach	2.27	2.38	2.37
	pickup	2.36	2.74	2.61
	carry	2.09	2.33	2.20
PSM	approach	2.17	2.23	2.44
	pickup	2.22	2.98	2.75
	carry	2.10	2.45	2.56
CRL	approach	2.50	3.02	2.71
	pickup	2.73	4.56	3.31
	carry	2.58	3.00	2.95
RLDP	approach	2.32	2.36	2.55
	pickup	2.64	3.06	3.11
	carry	2.22	2.65	2.80

A.4 Backward-embedding geometry probe

Table 5: Carry-regime resolution of the backward embedding B for cube-to-goal distance d, under a linear and a nonlinear (MLP) readout, with the maintain-control failure rate. This probe is restricted to methods with an explicit backward embedding. Better cube-goal geometric resolution in B does not imply better post-contact control.

Method	$R^2(B \to d)$ linear	$R^2(B \to d)$ MLP	Maintain-control fail (%)
FB	0.20	0.21	24.9
RLDP	0.55	0.75	43.8
PSM	0.90	**0.90**	36.4

A.5 Counterfactual value and representation ranking

The main value-quality diagnostic uses an on-policy counterfactual label. We reset the simulator to in-hand grasp states, roll out each agent's own policy, and label each state by whether that agent reaches the goal. This directly tests whether the value the policy follows ranks post-acquisition states by their eventual tail outcome.

Table 6: On-policy counterfactual value test. We reset 200 in-hand grasps into the simulator, roll out each agent's own policy, and report the per-task rank-AUC of the learned value against the agent's own success, averaged over goals, along with mean counterfactual success rate. This is the value-quality measure used in the main text.

Method	Counterfactual value-AUC	Counterfactual success rate
GCIQL	0.89	0.66
CRL	0.68	0.38
FB	0.61	0.55
RLDP	0.52	0.21
PSM	0.49	0.32

Figure 5 compares the learned value against a supervised linear readout from the representation that the value uses. This asks whether the representation contains linearly accessible information about the agent's own counterfactual success, and whether the learned scalar value preserves that information in its ranking.

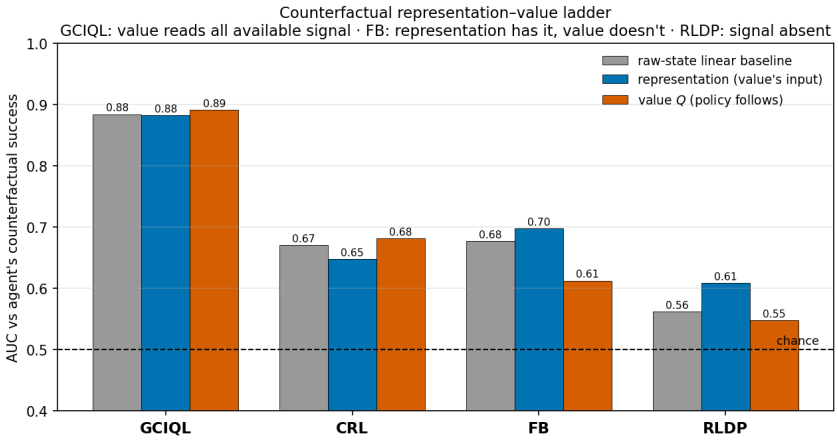


Figure 5: Counterfactual representation–value ladder. For each method, the probe uses the agent's own counterfactual success as the label. We compare a raw-state linear baseline, the best linear readout of the representation used by the value (GCIQL penultimate value features; CRL $\phi(s, \pi)$; FB/RLDP forward map $F(s, \pi, z)$), and the learned value's own ranking. GCIQL preserves nearly all linearly decodable signal available in its value features; FB contains more signal in F than is reflected in the learned scalar value; RLDP contains weaker signal under this probe. The raw baseline is per-agent because counterfactual success depends on each policy, so the cleanest comparison is within method.